# LABORATORY OF AUTOMATION SYSTEMS

Davide Samorì

Dipartimento di Ingegneria dell’Energia Elettrica e dell’Informazione (DEI)

Università di Bologna

Email: davide.samori@unibo.it

TODAY GOAL:

- Matlab/Simulink Review
- DC Motor model
  - Equation from Prof.Melchiorri slide
  - Motor Ratings
  - Compare model results with motor ratings
- Graph in Simulink/Matlab
- Time Constant

**PC LAB5**

Username: **Studente**
Password: **DEI01**

Using Laplace transformation, it results:

[1] $$\boxed{V_A(s) = R_A I_A(s) + sL_A I_A(s) + V_M(s)}$$

[2] $$C_m(s) = k_m \; I_A(s)$$

[3] $$\boxed{s \; J \; \Omega_m(s) = C_m(s) - b \; \Omega_m(s)}$$

[4] $$V_M(s) = k_m \; \Omega_m(s)$$

The two dynamics are then expressed by the transfer functions:

[5] $$I_A(s) = \frac{V_A(s) - V_M(s)}{R_A + s \; L_A}$$

[6] $$\frac{\Omega_m(s)}{C(s)} = \frac{1}{b + s \; J}$$

Slide Prof. Melchiorri
LAS_03_Motion.pdf
Pag. 10

$$\cancel{C(s)}\ \frac{\Omega_m(s)}{\cancel{C(s)}} = \frac{1}{b + s\ J}\ C(s)$$

$$\Omega_m(s) = \frac{C(s)}{b + sJ}$$

$$I_A(s) = \frac{V_A(s) - V_M(s)}{R_A + s\ L_A}$$

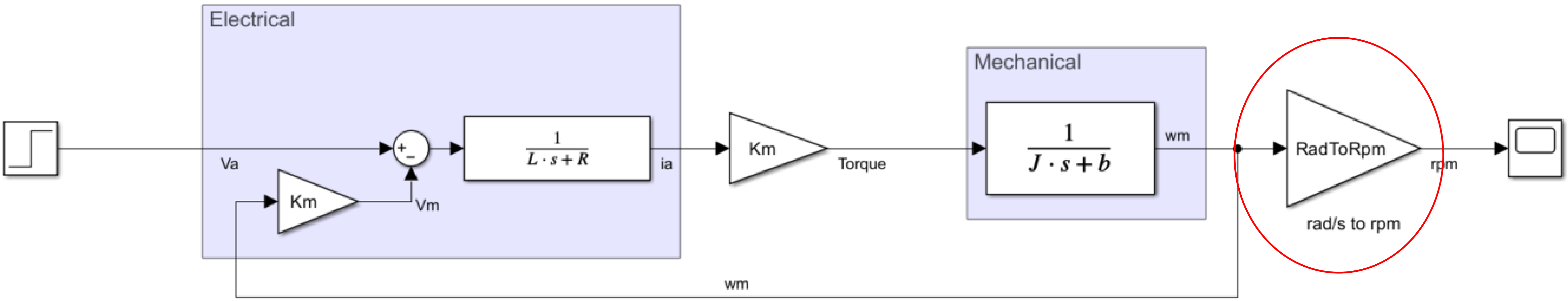
$$= I_a(s) \cdot k_m \cdot \frac{1}{b + sJ}$$

$$C_m(s) = k_m\ I_A(s)$$

$$= \frac{V_A(s) - V_M(s)}{R_A + sL_A} \cdot k_m \cdot \frac{1}{b + sJ}$$

$$V_M(s) = k_m\ \Omega_m(s)$$

$$\Omega_m(s) = \left(V_A(s) - k_m \cdot \Omega_m(s)\right) \cdot \frac{1}{R_A + sL_A} \cdot k_m \cdot \frac{1}{b + sJ}$$




# EXERCISE

Let's now create a parameter file:

**Matlab -> new Script**

```
%% DC MOTOR PARAMETER

L =        ?   ;      % Inductance (H)
R =        ?   ;      % Resistance (Ohm)
Km =       ?   ;      % Torque constant (Nm/A)
J =        ?   ;      % Rotor inertia (kg m2)
b=         ?   ;      % Coefficient of friction

%% CONVERSION
RadToRpm= ? ;
```

## Try to keep all parameters in order

- Pay attention to the units of measurement
- If you change motor, you have just to edit a few lines of text
- If you want to experiment with changing a parameter, you only have to change one number, you don't have to scroll the entire model
- The model is much more readable if another person has to look at it

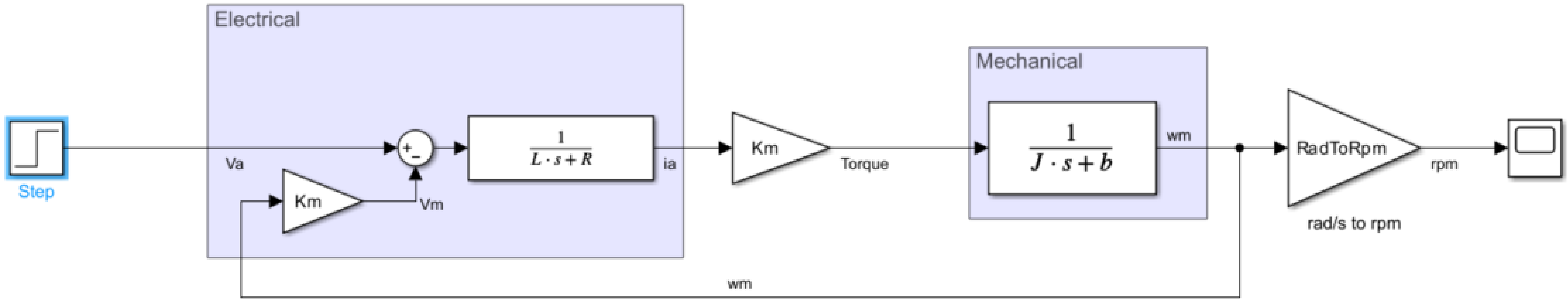
# EXERCISE

| DATI MOTORE<br>Motor ratings | SIMBOLI<br>Symbols | UNITA'<br>Units | PMB 1524 |
|---|---|---|---|
| COPPIA ALLA VELOCITA' NOMINALE Torque at rated speed | Cn | Nm | 0.22 |
| VELOCITA' NOMINALE Rated speed | Nn | RPM | 1500 |
| POTENZA NOMINALE Rated output | Pu | W | 35 |
| TENSIONE NOMINALE Rated voltage | Vn | V | 24 |
| CORRENTE NOMINALE Rated current | In | A | 1.9 |
| VELOCITA' A VUOTO Idle speed | Nv | RPM | 1750 |
| TENSIONE A VUOTO Idle voltage | Vv | V | 24 |
| CORRENTE A VUOTO Idle current | Iv | A | 0.3 |
| F.C.E.M ALLA VELOCITA' NOMINALE B.E.M.F at rated speed | E | V | 19.5 |
| COPPIA DI PICCO Peak torque | Cp | Nm | 0.88 |
| CORRENTE DI PICCO Peak current | Ip | A | 8 |
| MAX VELOCITA' Max speed | N max | RPM | 4000 |
| RENDIMENTO Efficiency | η | % | 73 |
| DATI MECCANICI Mechanical data | | | |
| MOMENTO D'INERZIA ROTORICO Rotor inertia | J | Kgcm² | 0.352 |
| MAX. ACCELLERAZ. TEORICA Max theorical accelleration | α | rad/sec² | 25000 |
| MAX TENSIONE Max voltage | V max | V | 65 |
| COSTANTE DI TEMPO MECCANICA Mechanical time constant | Tm | ms | 85 |
| VENTILAZIONE Ventilation | | | Naturale |
| GRADO DI PROTEZIONE Protection (IEC 34.5) | | IP | 54 |
| PESO Weight | G | Kg | 1.35 |
| DATI ELETTRICI Electric data | | | |
| COSTANTE DI TENSIONE Voltage constant | Ke | Vs\rad | 0.12 |
| COSTANTE DI COPPIA Torque constant | Kt | Nm\A | 0.11 |
| COSTANTE DI TEMPO TERMICA Thermal time constant | Tt | min | 15 |
| COSTANTE DI TEMPO ELETT. Electrical time constant | Te | ms | 1.3 |
| RESISTENZA D'ARMATURA Armature resistance | Ra | Ohm | -- |
| RESISTENZA D'ARMATURA CON SPAZZOLE Terminal resistance | Rm | Ohm | 3.20 |
| INDUTTANZA D'ARMATURA Armature inductance | La | mH | 4.1 |

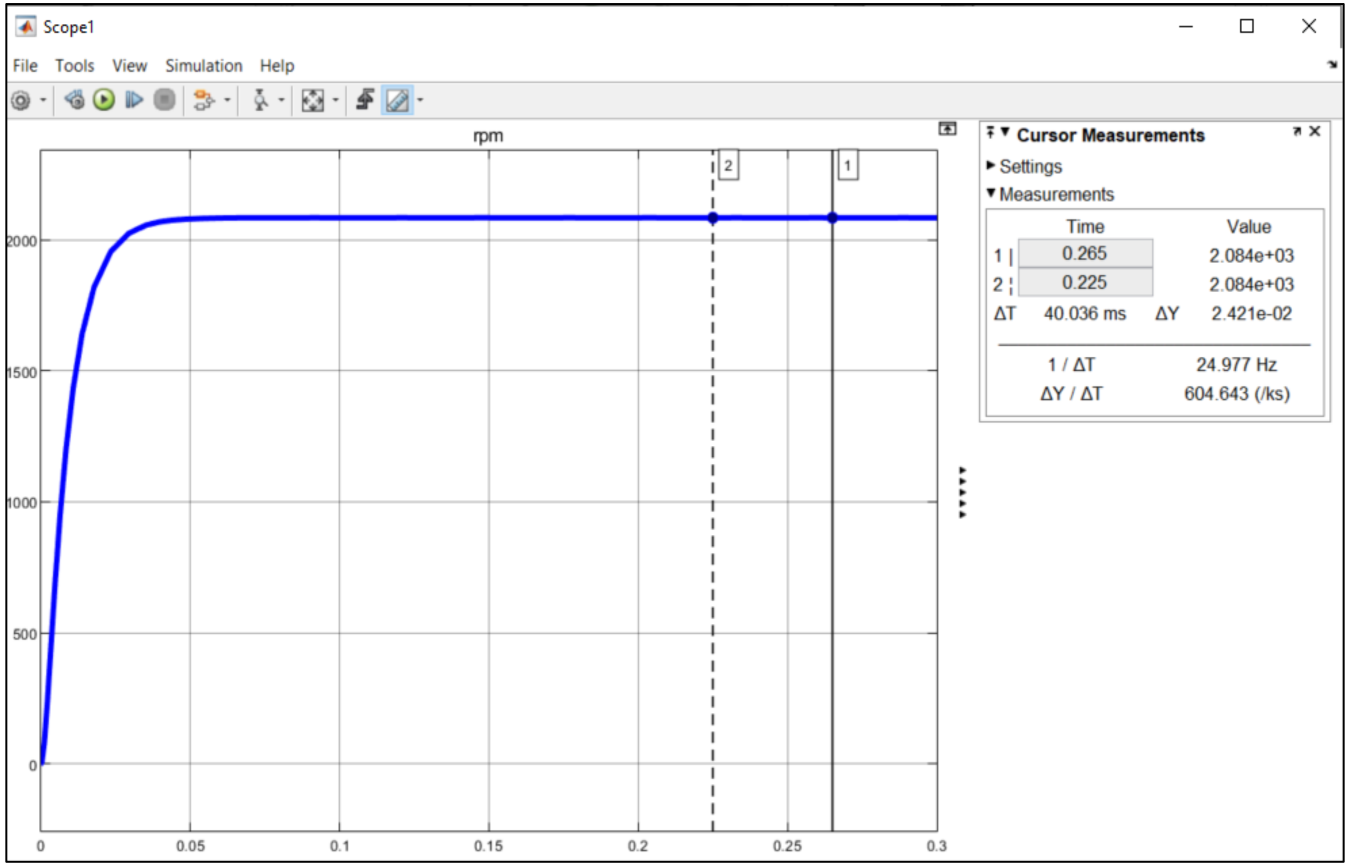
**you should be able to derive the highlighted values**

You can try to get as more parameters you can.

| | | | |
|---|---|---|---|
| VELOCITA' A VUOTO    Idle speed | Nv | RPM | 1750 |



| VELOCITA' A VUOTO Idle speed | Nv | RPM | 1750 |
|---|---|---|---|



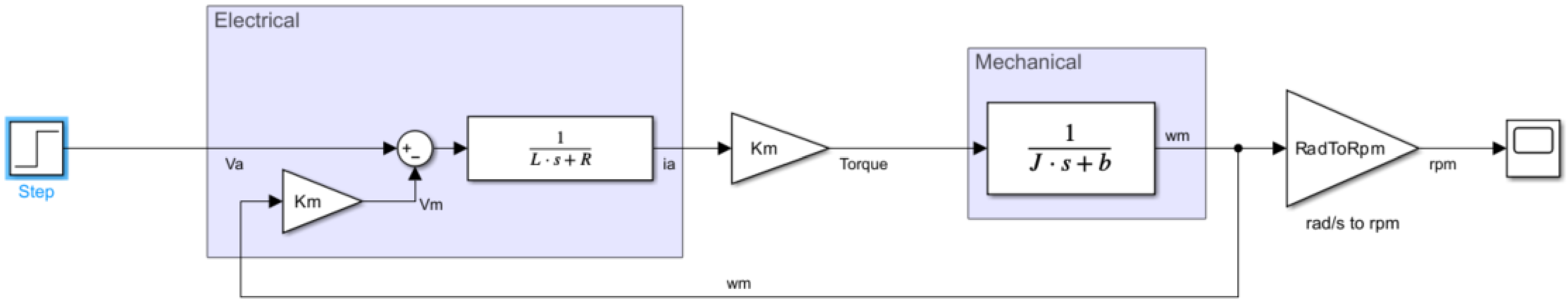

Omega=2048rpm

higher but we have neglected friction as it is not indicated in the datasheet

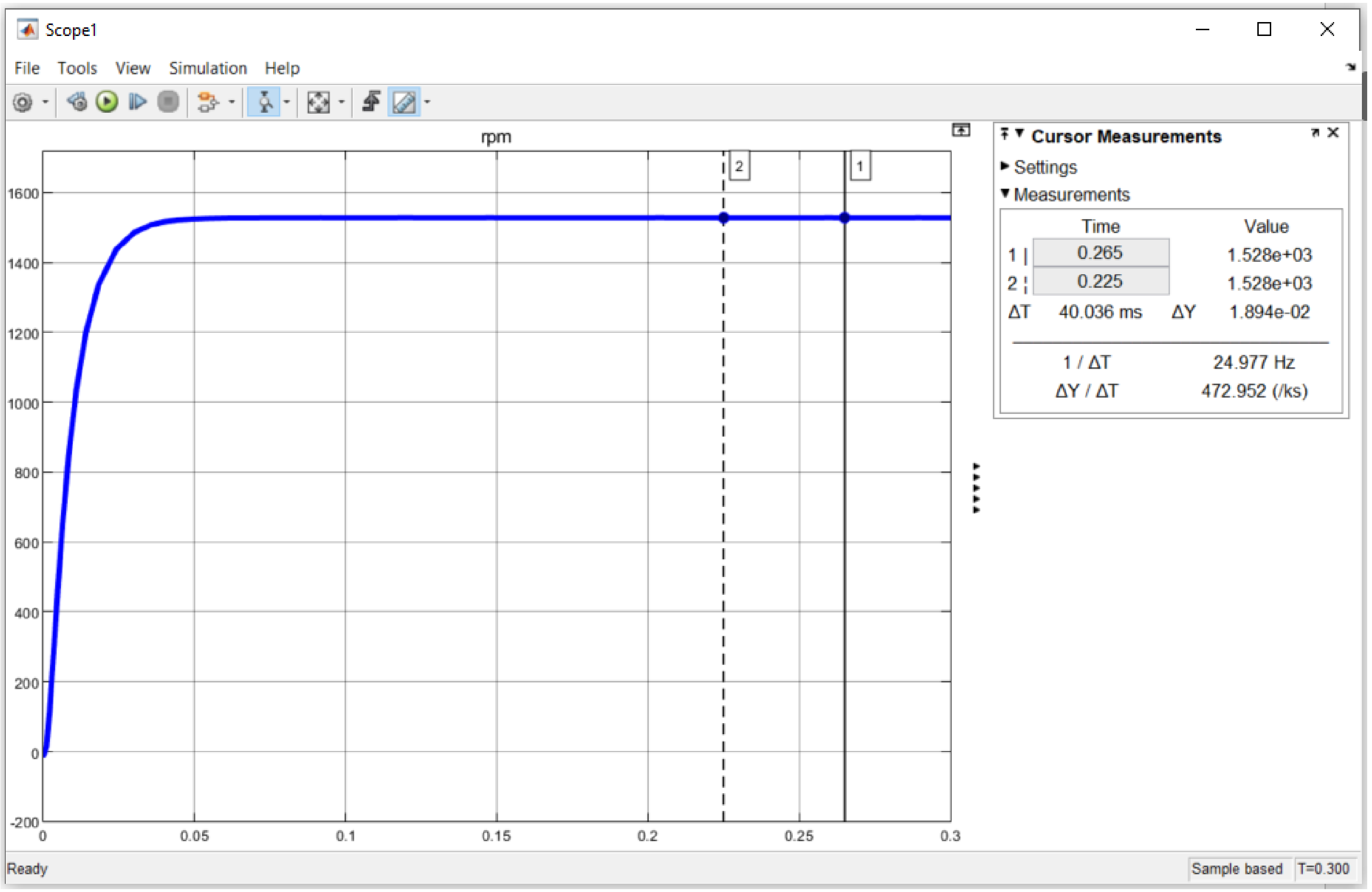
What can we do?

| | | | |
|---|---|---|---|
| VELOCITA' NOMINALE Rated speed | Nn | RPM | 1500 |

Speed with nominal Torque, we have to add something…



| | | | |
|---|---|---|---|
| VELOCITA' NOMINALE Rated speed | Nn | RPM | 1500 |

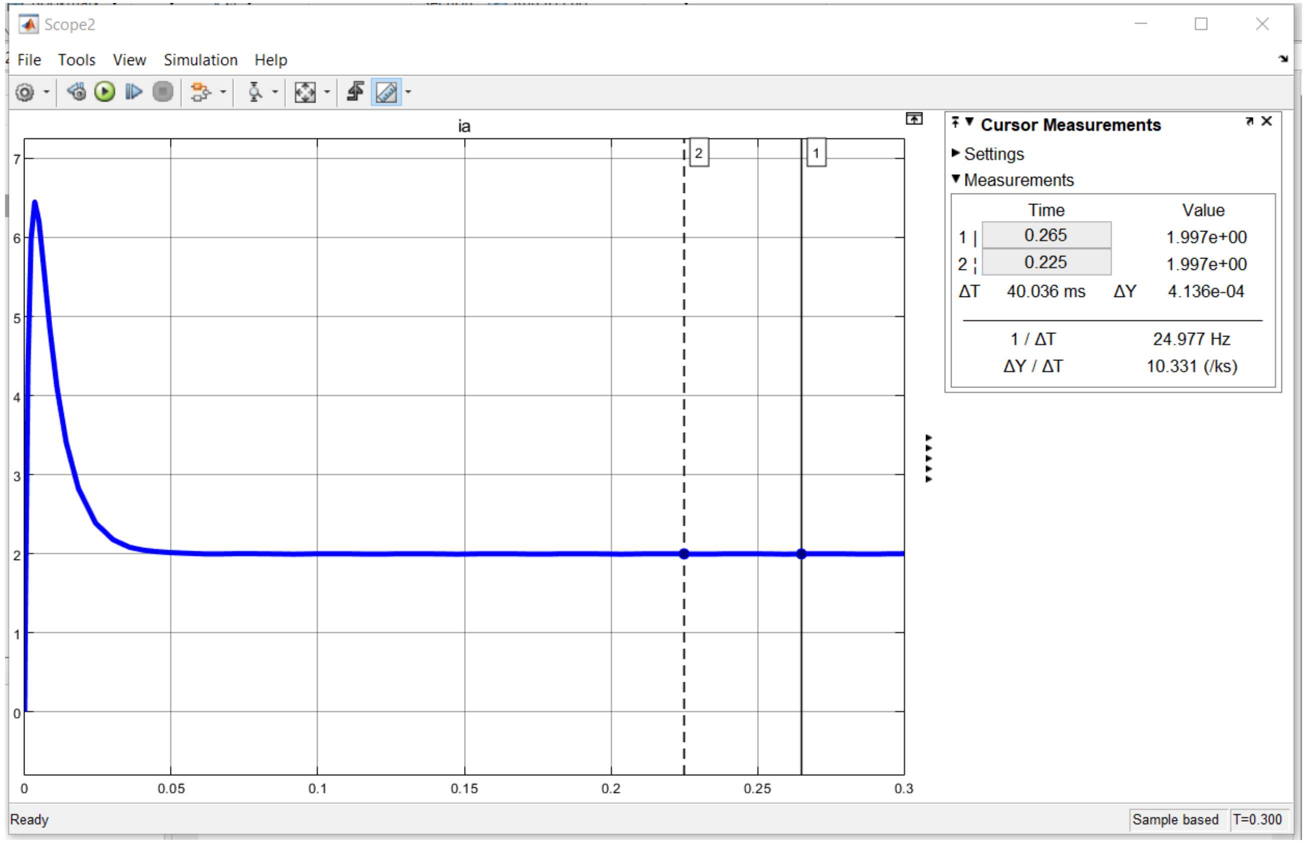



very close

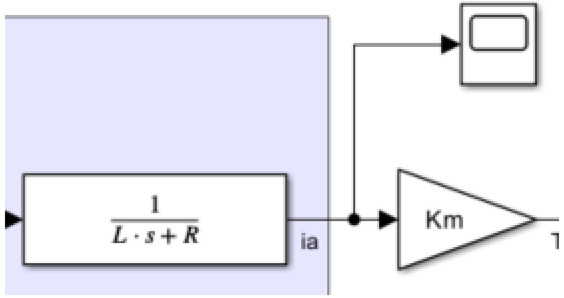
| | | | |
|---|---|---|---|
| VELOCITA' NOMINALE Rated speed | Nn | RPM | 1500 |

We can also check Rated current **In**:

| | | | |
|---|---|---|---|
| CORRENTE NOMINALE Rated current | In | A | 1.9 |




very close



Something more difficult…

| DATI ELETTRICI Electric data | | | |
|---|---|---|---|
| COSTANTE DI TENSIONE Voltage constant | Ke | Vs\rad | 0.12 |
| COSTANTE DI COPPIA Torque constant | Kt | Nm\A | 0.11 |
| COSTANTE DI TEMPO TERMICA Thermal time constant | Tt | min | 15 |
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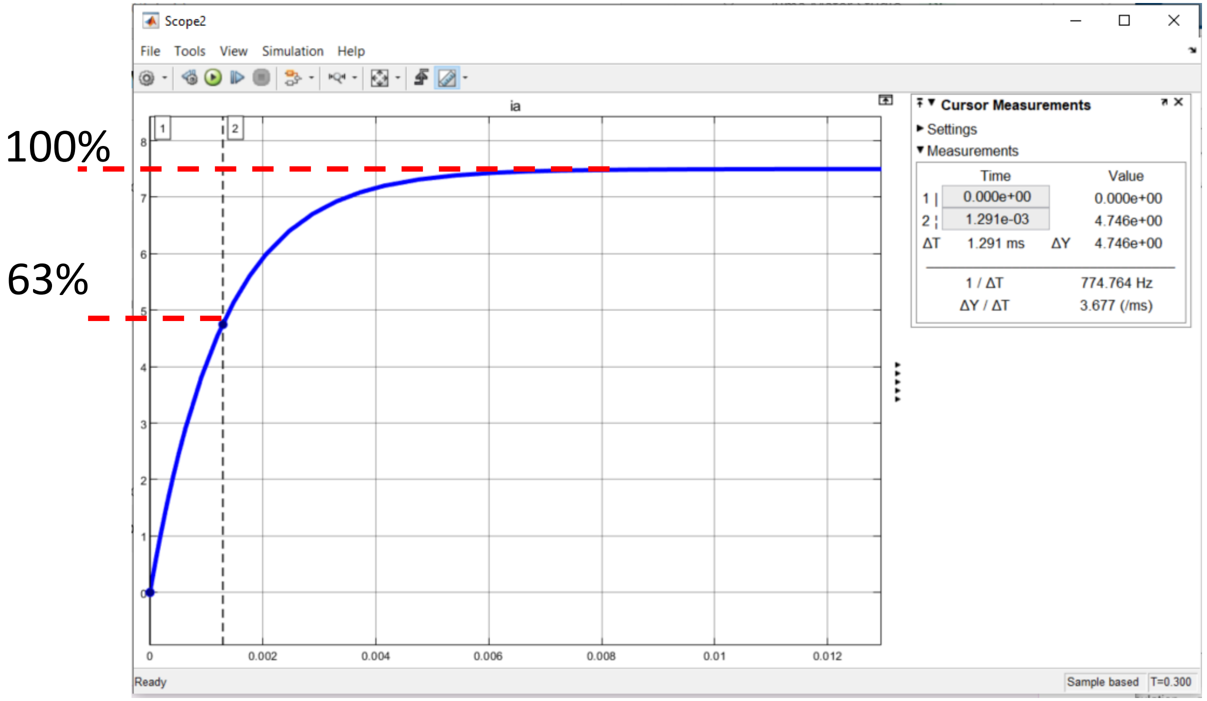
| RESISTENZA D'ARMATURA Armature resistance | Ra | Ohm | -- |
| RESISTENZA D'ARMATURA CON SPAZZOLE Terminal resistance | Rm | Ohm | 3.20 |
| INDUTTANZA D'ARMATURA Armature inductance | La | mH | 4.1 |

## Electrical time constant?




How can you get it?

Te=1.3ms

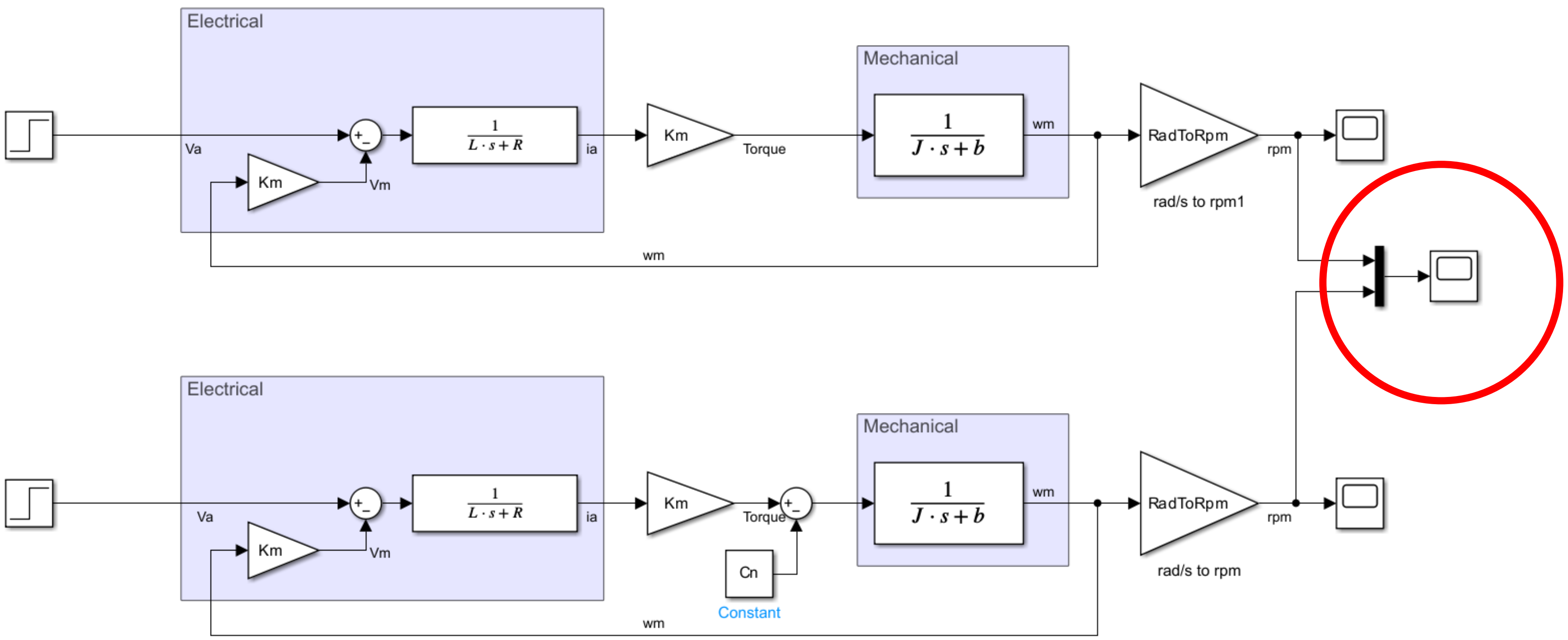
## WORK WITH ACQUIRED DATA

It is often useful to work with previously acquired data:

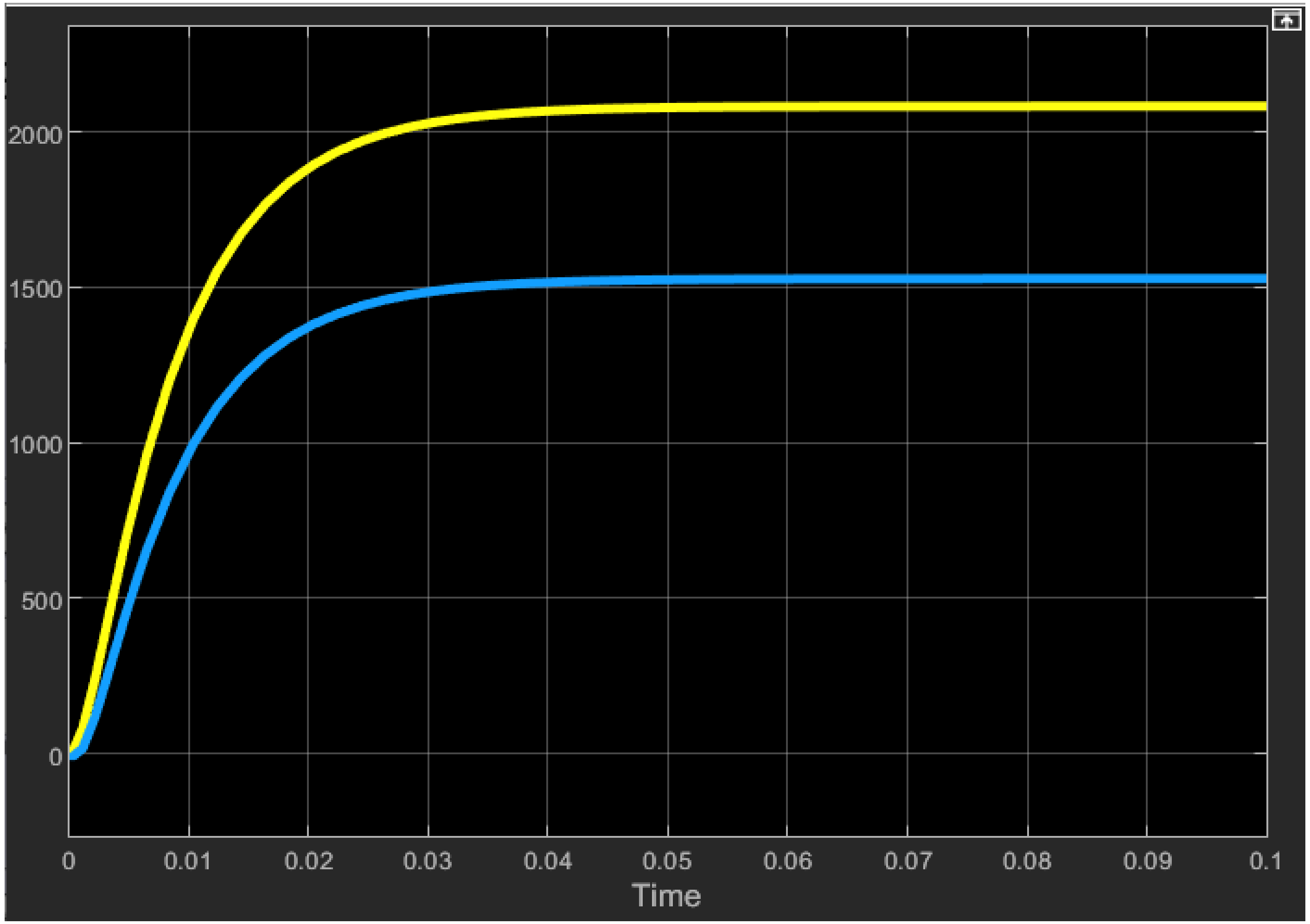
- compare different control strategy
  - In the setup we can only run one control algorithm at a time
  - In Simulink you can run two or more control algorithm
- Post data processing (Eg. Time constant, overshoot, ecc…)
- Compare real data with simulation data
- Ecc…

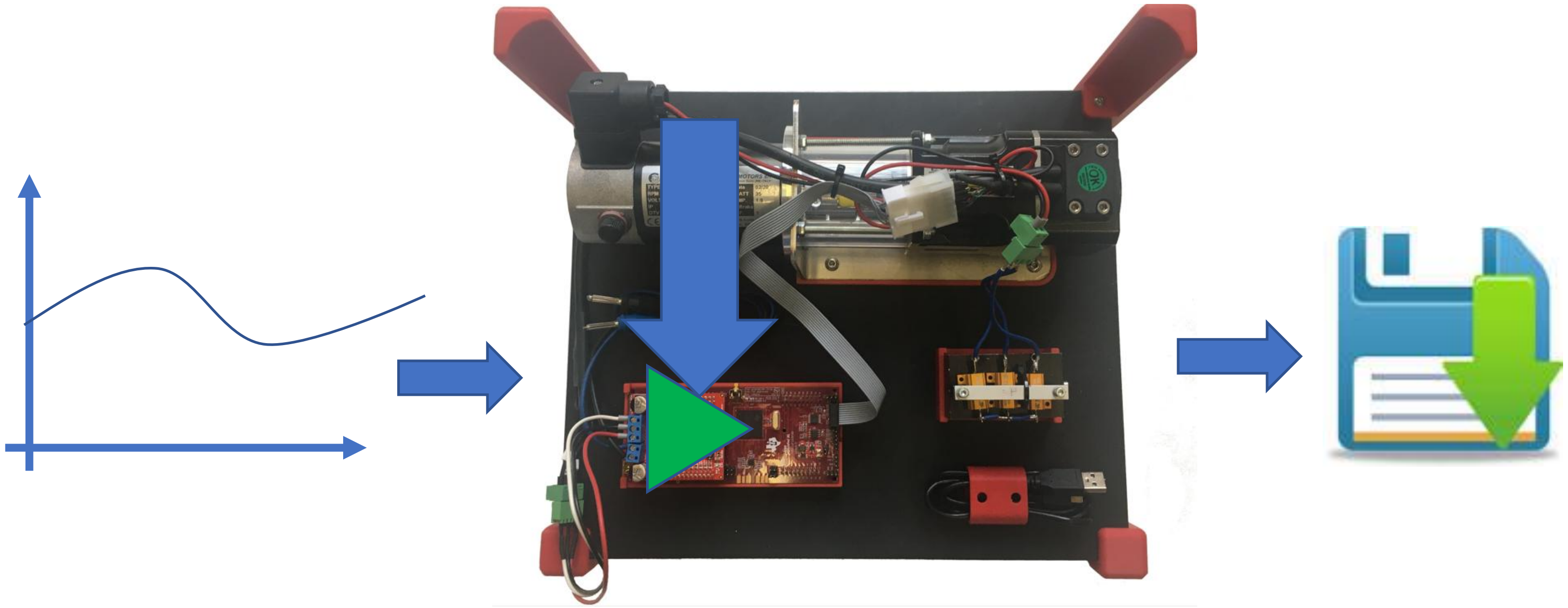
WORK WITH ACQUIRED DATA

Suppose we want to compare the speed of our motor (model) with or without load:



# WORK WITH ACQUIRED DATA





We can't duplicate it!

IF WE WANT COMPARE TWO CONTROL SYSTEMS:

1- DOWNLOAD 1° CONTROL

2- RUN THE CONTROL

3- GIVE A REFERENCE

4- ACQUIRE DATA

1- DOWNLOAD 2° CONTROL

2- RUN THE CONTROL

3- GIVE A REFERENCE

4- ACQUIRE DATA

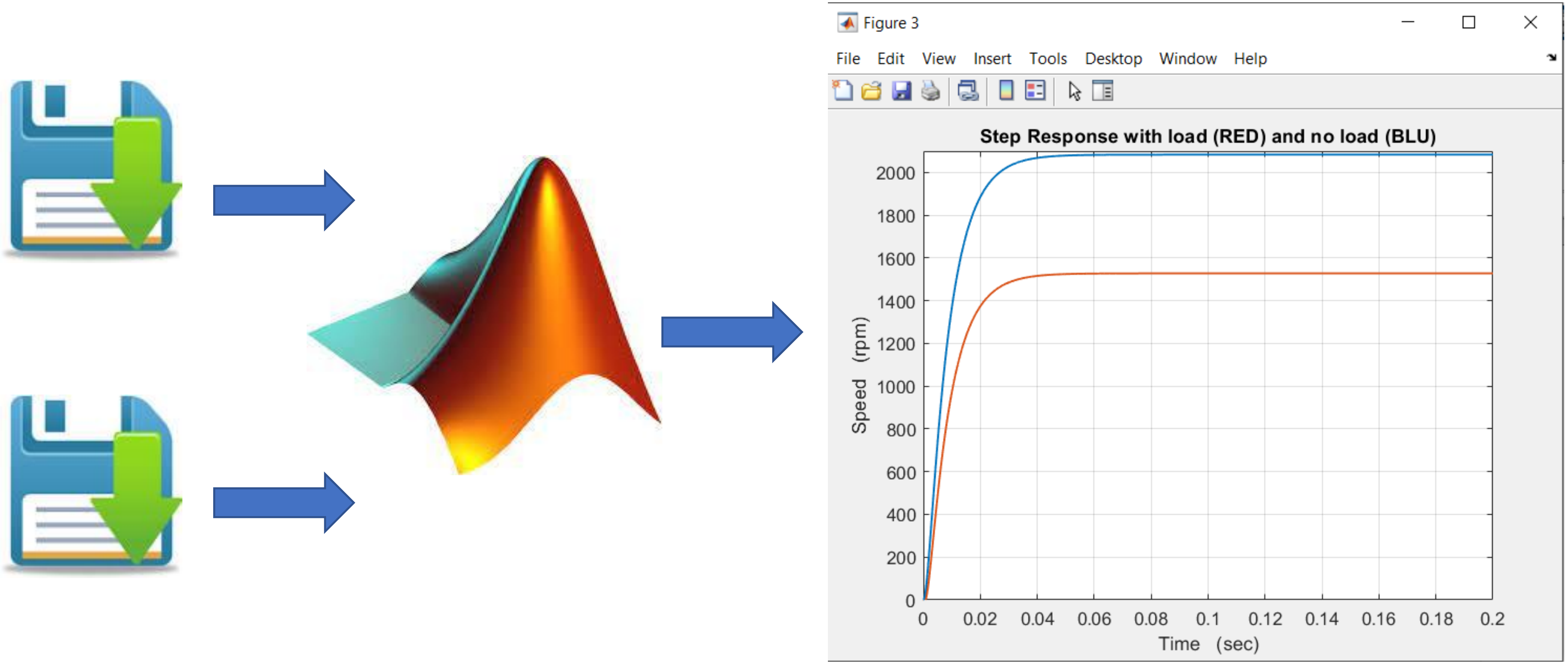
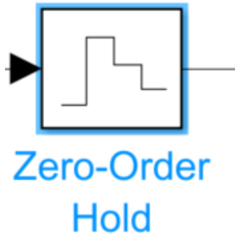


Figure 3
File Edit View Insert Tools Desktop Window Help
Step Response with load (RED) and no load (BLU)
Speed (rpm)
Time (sec)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2

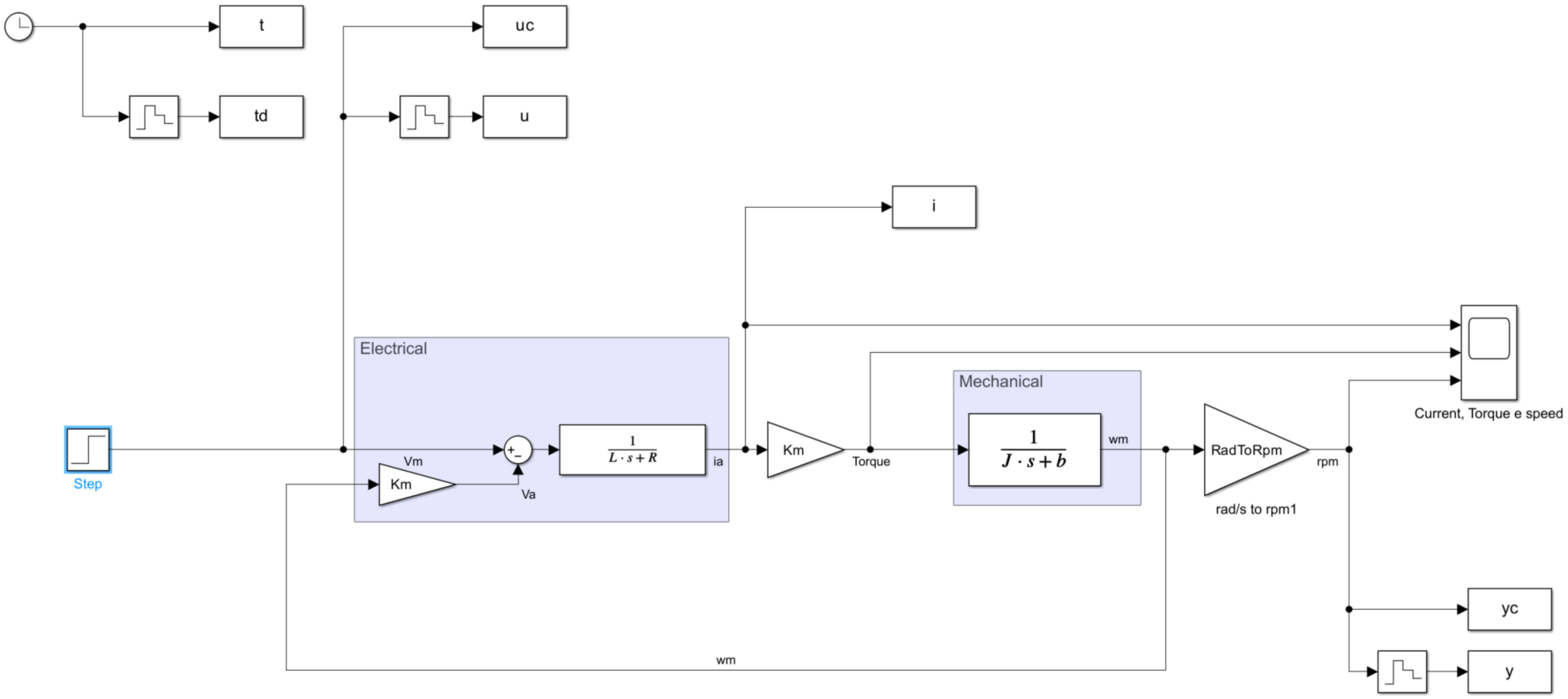
We are going to use:

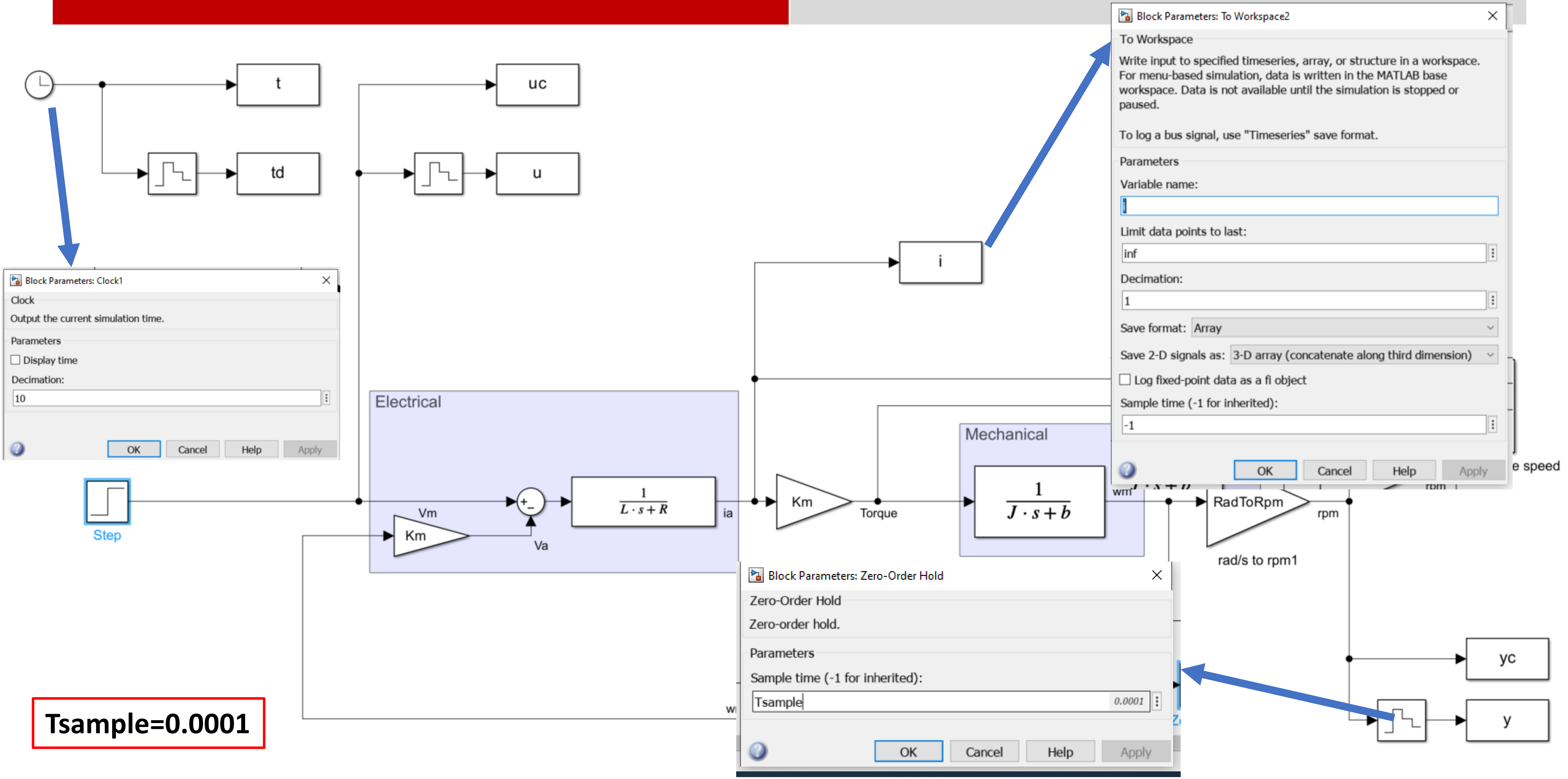

**Description**
The To Workspace block logs the data connected to its input port to a workspace from a Simulink® model.




**Description**
The Zero-Order Hold block holds its input for the sample period you specify. If the input is a vector, the block holds all elements of the vector for the same sample period.


**Description**
The Clock block outputs the current simulation time at each simulation step. This block is useful for other blocks that need the simulation time.


t
td
uc
u
i
Electrical
Mechanical
Step
Vm
Km
Va
$\frac{1}{L \cdot s + R}$
ia
Km
Torque
$\frac{1}{J \cdot s + b}$
wm
RadToRpm
rad/s to rpm1
rpm
Current, Torque e speed
yc
y
wm



t
td
uc
u
i
Block Parameters: Clock1
Clock
Output the current simulation time.
Parameters
Display time
Decimation:
10
OK
Cancel
Help
Apply
Block Parameters: To Workspace2
To Workspace
Write input to specified timeseries, array, or structure in a workspace. For menu-based simulation, data is written in the MATLAB base workspace. Data is not available until the simulation is stopped or paused.
To log a bus signal, use "Timeseries" save format.
Parameters
Variable name:
Limit data points to last:
inf
Decimation:
1
Save format: Array
Save 2-D signals as: 3-D array (concatenate along third dimension)
Log fixed-point data as a fi object
Sample time (-1 for inherited):
-1
OK
Cancel
Help
Apply
Electrical
Mechanical
Vm
Km
Va
$\frac{1}{L \cdot s + R}$
ia
Km
Torque
$\frac{1}{J \cdot s + b}$
wm
RadToRpm
rpm
rad/s to rpm1
Step
Block Parameters: Zero-Order Hold
Zero-Order Hold
Zero-order hold.
Parameters
Sample time (-1 for inherited):
Tsample
0.0001
OK
Cancel
Help
Apply
yc
y
Tsample=0.0001

## MATLAB CODE:

```
%% simulation
LW = 1.1;                %Line Width
MaxAxisRPM=2100;  %YMax Graph

Ttot = 0.2;              %Simulation time
sim('DC_Motor.slx'); %Run model to collect data

%% Plot 24V input step response
figure
PP = plot(td,y);
set(PP,'LineWidth',LW)
grid
axis([0 Ttot 0 MaxAxisRPM])
xlabel('Time (sec)')              %ALWAYS UNIT
ylabel('Speed (rpm)')             %ALWAYS UNIT
title('Motor velocity with 24V Step input') %ALWAYS TITLE
```

```matlab
%% check time constant
ymax = max(yc);
tmax=max(td);
yt = ymax*0.6321;
ydiff = abs(yc - yt);
[mm,yy] = min(ydiff);

tau_y = td(yy)
gain_y = ymax;

% parameters
numi = gain_y;
deni = [tau_y, 1];
gi = tf(numi,deni);

% simulation of electric dynamics
[yr,tc,] = step(gi,tmax);
```

```matlab
% comparison
figure
plot(tc,yr,'LineWidth',LW)
hold on
plot(t,yc,'LineWidth',LW)
hold off
grid
axis([0 tmax 0 MaxAxisRPM])
xlabel('Time (sec)')
ylabel('Speed (rpm)')
title(['First Order System approximation - TimeConstant=',num2str(tau_y),'s'])
```

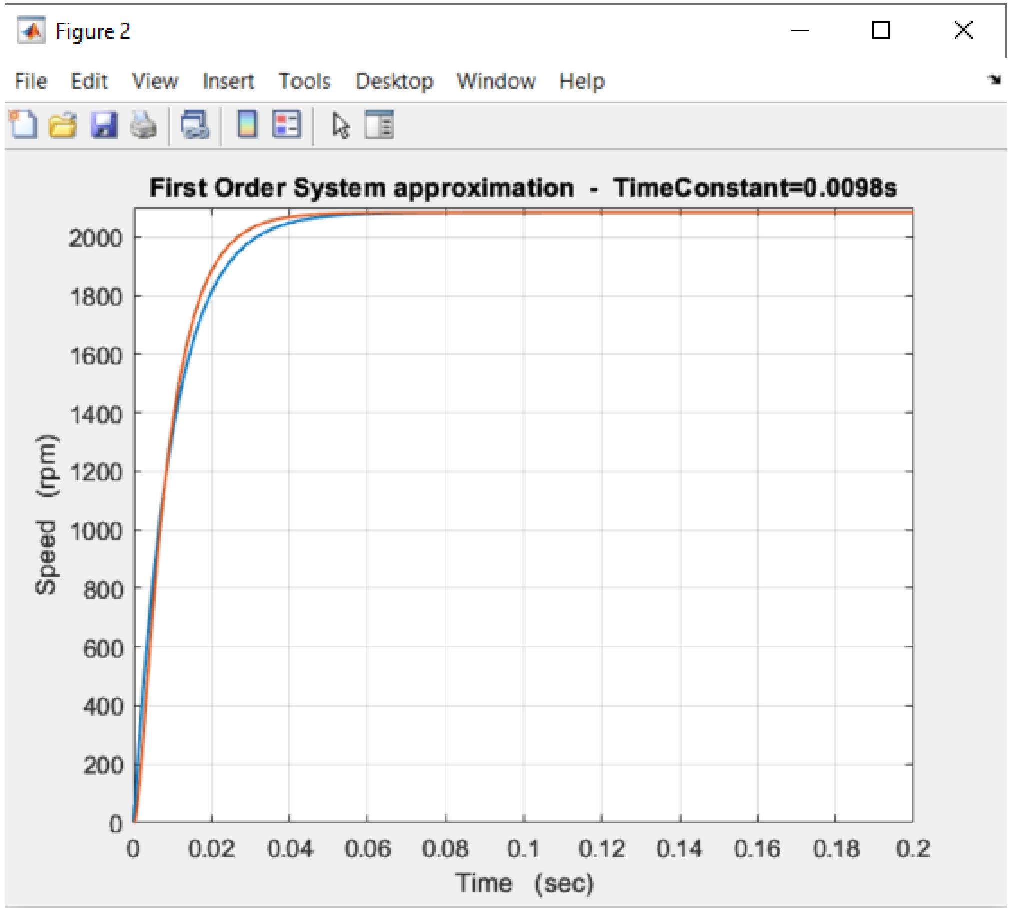


Figure 2
File Edit View Insert Tools Desktop Window Help
First Order System approximation - TimeConstant=0.0098s
Speed (rpm)
Time (sec)

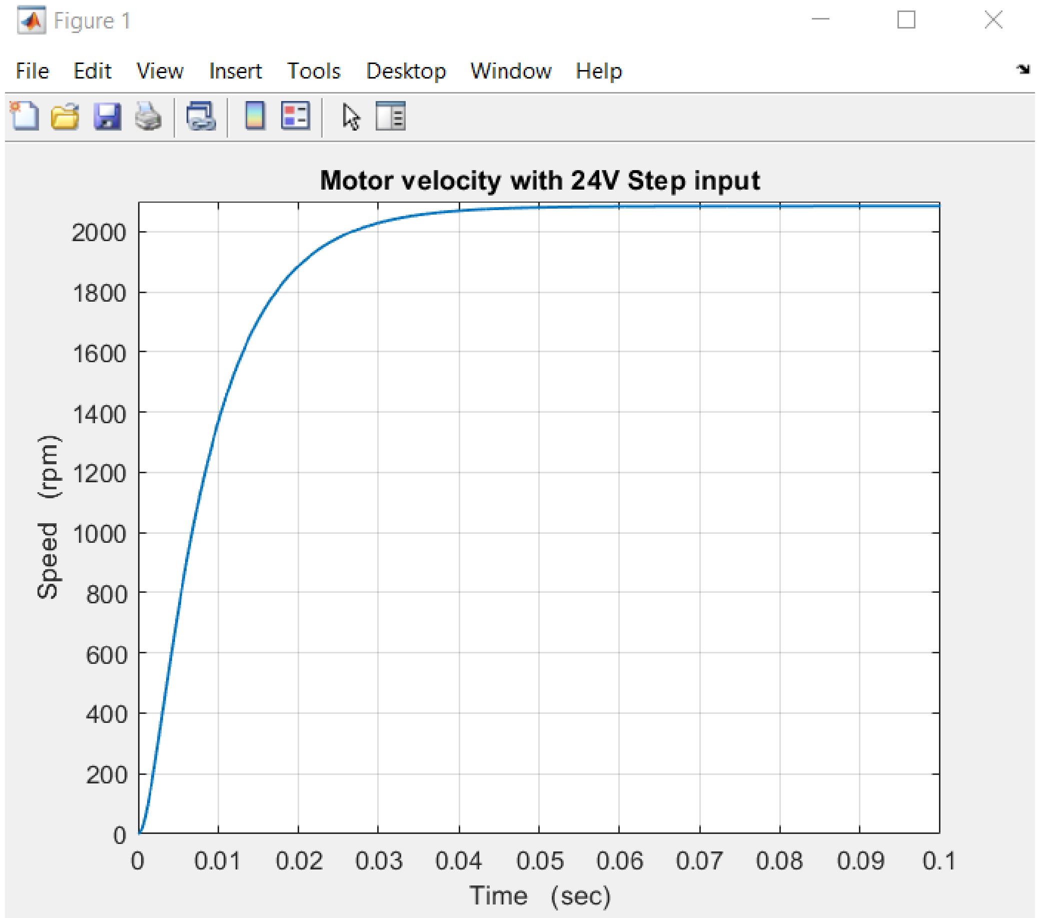
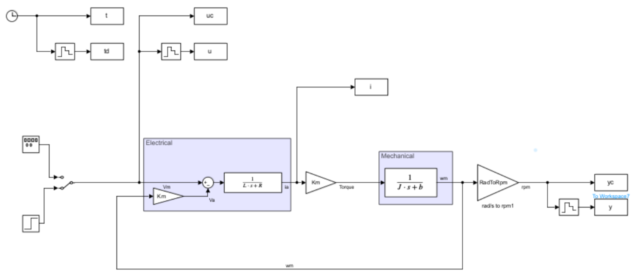


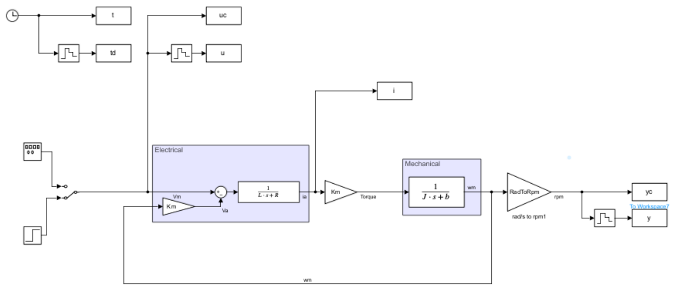
Figure 1
File Edit View Insert Tools Desktop Window Help
Motor velocity with 24V Step input
Speed (rpm)
Time (sec)

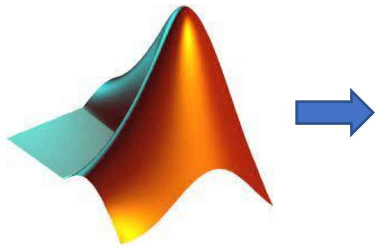
1. Run a simulation with no load
2. Store Data
3. Run a simulation with load
4. Store Data

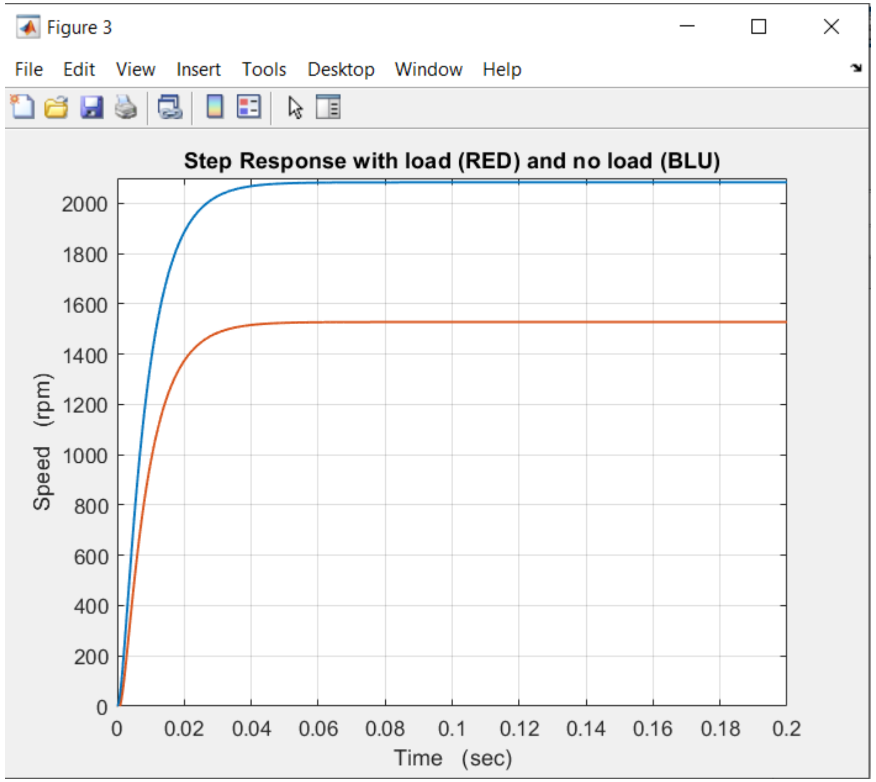
5. Plot both data in the same matlab graph

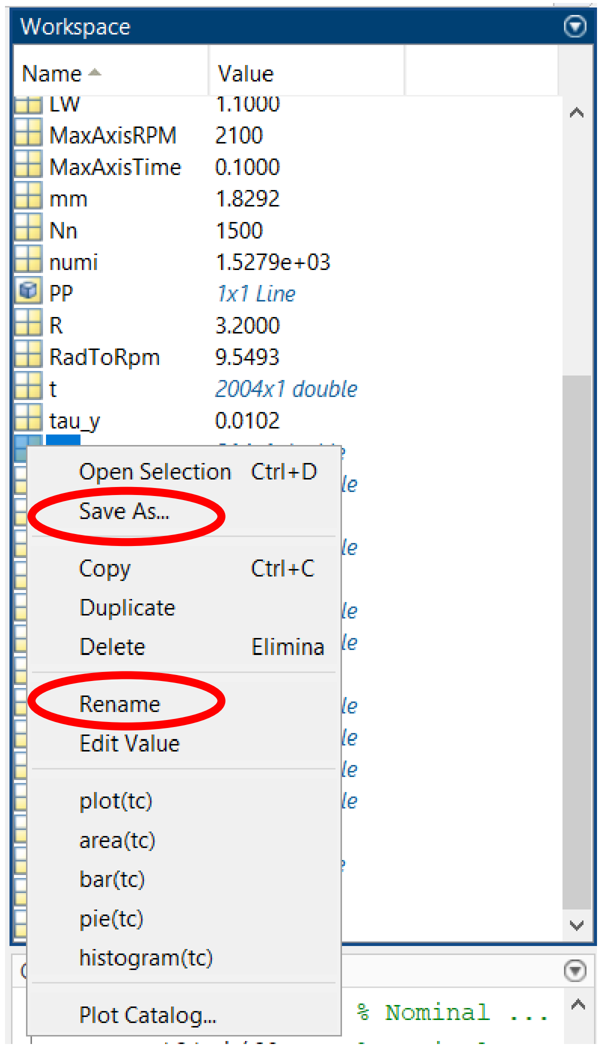
**Home exercise**












TIP:

Right-click on vector

- you can save values into .mat file
- you can rename it

You can lose your workspace with a clean command or shutdowm